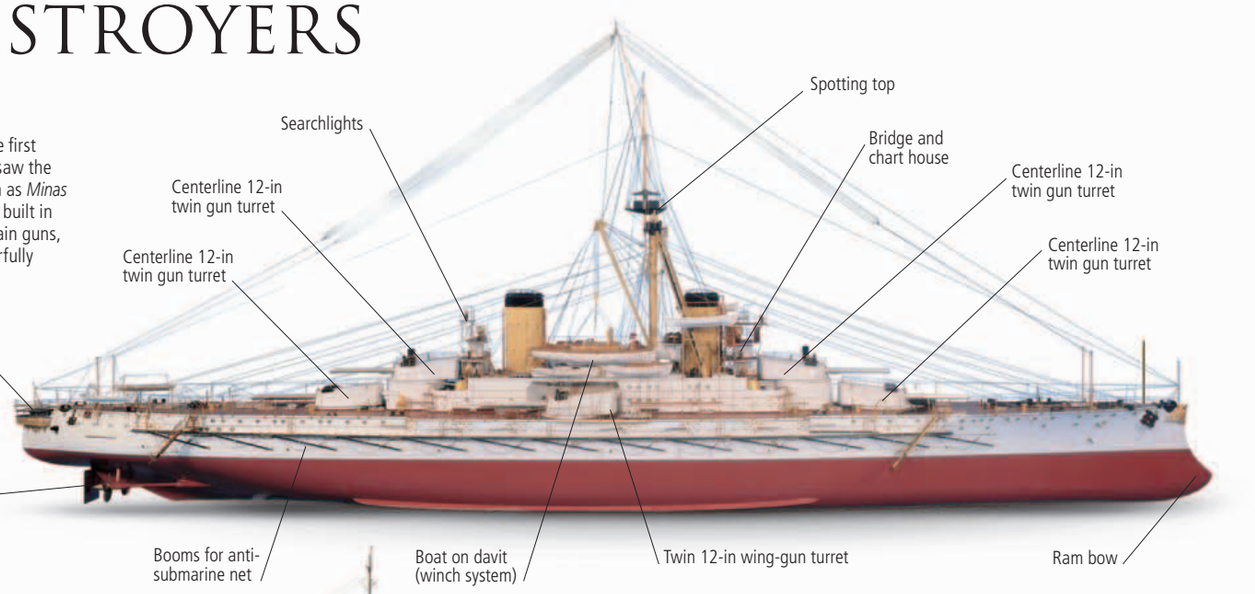
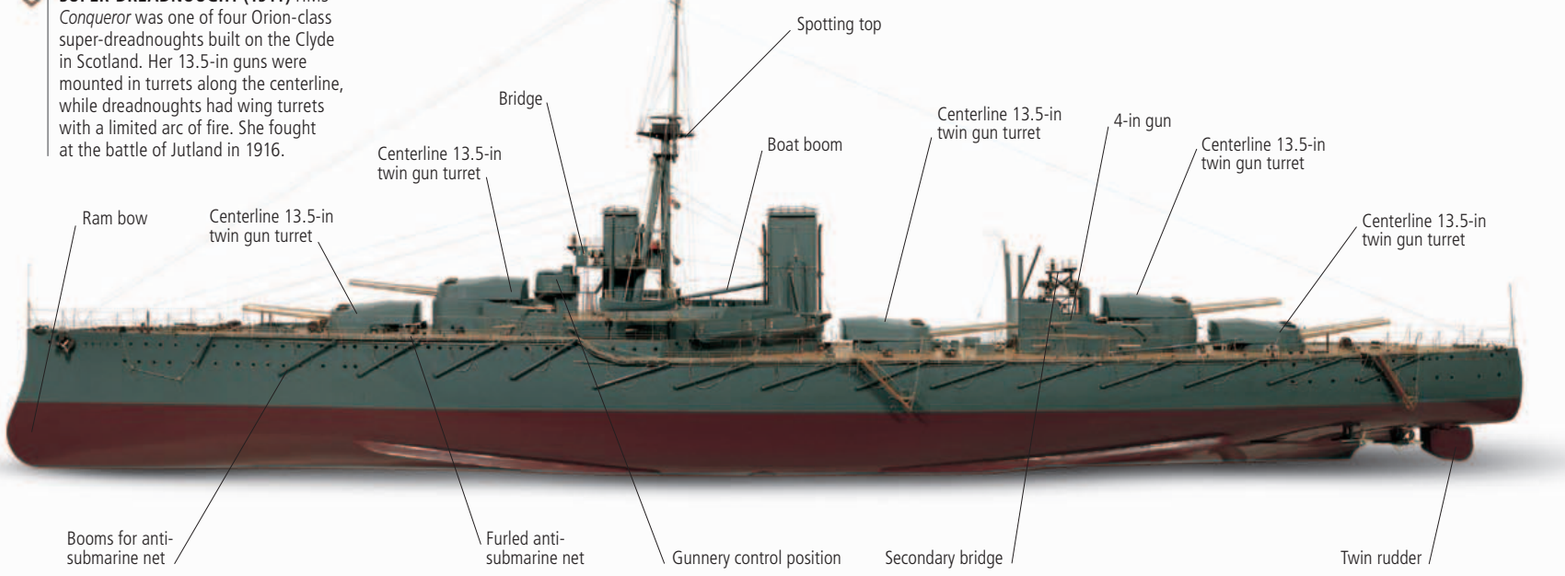
DREADNOUGHTS, CRUISERS, AND DESTROYERS

DREADNOUGHT (1910) The first decade of the 20th century saw the arrival of dreadnoughts such as *Minas Gerais*, a Brazilian battleship built in Britain. Carrying 12 12-in main guns, she was one the most powerfully armed ships of the time.



SUPER-DREADNOUGHT (1911) HMS *Conqueror* was one of four Orion-class super-dreadnoughts built on the Clyde in Scotland. Her 13.5-in guns were mounted in turrets along the centerline, while dreadnoughts had wing turrets with a limited arc of fire. She fought at the battle of Jutland in 1916.



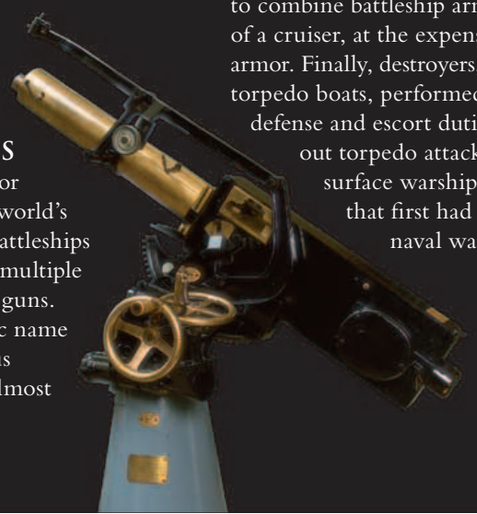
In the years between 1880 and the beginning of World War I in 1914, warship design settled into a generally recognizable format. Multiple guns were mounted along the centerline, and moving turrets provided wide arcs of fire. Armor plate, particularly around the magazine and the engine rooms, became critical, especially as guns, shells, propellants, and targeting equipment all went through dramatic improvements. By the end of World War I, battleships were sending shells of large caliber (12-in or even 15-in) accurately out to ranges of up to 60,000 ft (18,000 m).

POWER FACTORS

Propulsion also went through important changes from the last decades of the 19th century. The use of steam turbines produced better power more smoothly delivered, and during the second decade of the 20th century the major powers switched from coal-fired engines to oil-burning variants. Oil not only burned cleaner, but could be transferred easily at sea, and could power ships for longer distances.

TYPES OF WARSHIPS

Three primary types of major surface vessel equipped the world's navies from around 1900. Battleships were the capital ships, with multiple batteries of turret-mounted guns. The dreadnoughts, a generic name derived from the eponymous British battleship of 1906, almost exclusively carried very large-caliber guns for long-range ship-to-ship duels. Cruisers were faster and had longer range and lighter armament than battleships. The battlecruiser was designed to combine battleship armament with the speed of a cruiser, at the expense of much reduced armor. Finally, destroyers, developed from early torpedo boats, performed such roles as fleet defense and escort duties, as well as carrying out torpedo attacks. The bane of all surface warships was the submarine that first had a dramatic effect on naval warfare in World War I.



QF 1-pounder gun
The British QF 1-pounder was used as an anti-torpedo boat weapon during World War I.